THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-N-25027

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PREPARING ACTIVITY: DLA-ES

CUSTODIANS: ARMY- AV NAVY- AS
AIR FORCE-11 DLA-

REVIEW: ES
USER: CR, MI
PROJECT NUMBER: 5310-1856

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

MILITARY SPECIFICATION SHEET

TITLE NUT, SELF-LOCKING, PLATE, ONE LUG, NON-METALLIC INSERT, STEEL, 125 KSI FTU, 250°F

(D) ENTIRE STANDARD REVISED

INCH-POUND

SPECIFICATION SHEET NUMBER

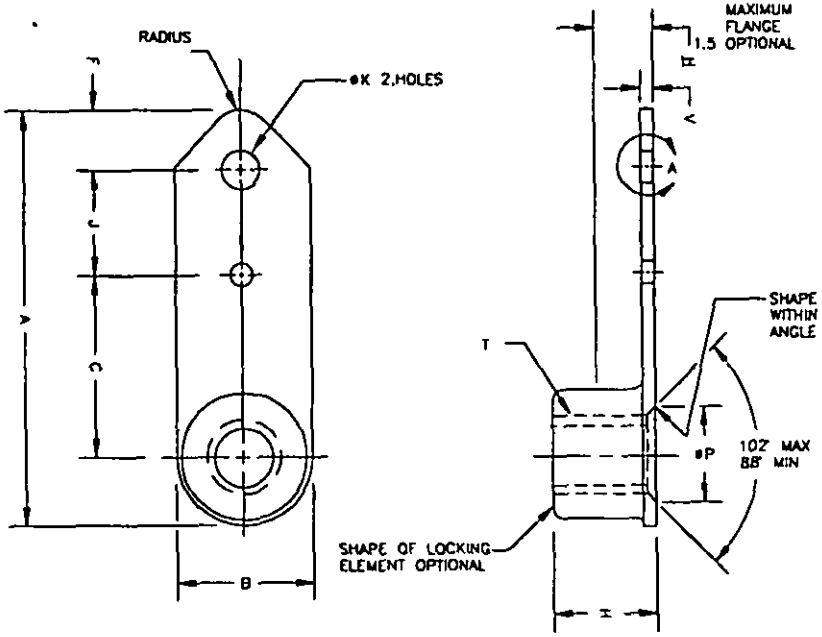
MS21080 D 20 APR 93

SUPERSEDING MS21080C 30 SEPT 87 (SEE NOTE 4)

AMSC- N/A FSC 5310

Page 1 of 2

CONFIGURATION OF NUT OPTIONAL WITHIN THE LIMITATIONS IMPOSED BY DIMENSIONS AND REQUIREMENTS AS SPECIFIED HEREIN



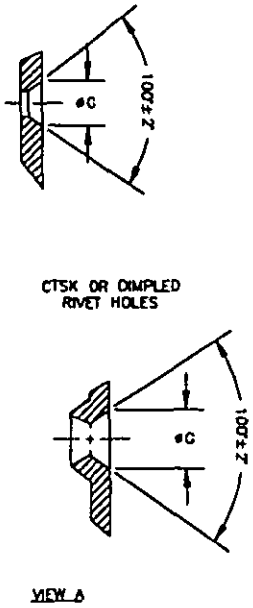


TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBERS		T	A	B		C	F	ØG	H		J 1/	ØK 2/	ØP		V	AXIAL STRENGTH	WEIGHT
PLAIN HOLES	CTSK OR DIMPLED HOLES	THREAD SIZES	MAX	MAX	MIN	±.005	MIN	±.010	MAX	MIN	±.002	±.005 -.000	MAX	MIN	MAX	LBS/MIN	LBS/100
-06	-06K	.138-32 UNJC-3B	1.031	.406	.375	.344	.115	.200	.297	.160	.312	.098	.177	.138	.075	1,130	1.40
-08	-08K	.164-32 UNJC-3B	1.031	.406	.375	.344	.115	.200	.297	.219	.312	.098	.230	.164	.075	1,720	1.30
-3	-3K	.190-32 UNJF-3B	1.031	.406	.375	.344	.115	.200	.312	.234	.312	.098	.230	.190	.075	2,460	1.20
-4	-4K	.250-28 UNJF-3B	1.062	.516	.438	.344	.115	.200	.375	.240	.312	.098	.293	.250	.075	4,580	1.50
-5	-5K	.3125-24 UNJF-3B	1.250	.531	.500	.500	.115	.230	.375	.271	.312	.130	.356	.312	.075	7,390	2.00
-6	-6K	.375-24 UNJF-3B	1.344	.641	.609	.500	.115	.230	.453	.318	.312	.130	.418	.375	.075	11,450	2.50

1/ THE CENTER OF THE TAPPED HOLE SHALL NOT DEVIATE IN ANY DIRECTION FROM THE CENTER OF THE PLATE NUT AS DETERMINED BY THE RIVET HOLES, BY MORE THAN .005.
2/ DIMPLED RIVET HOLE TOLERANCE FOR "ØK" IS +.015 -.000.

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REQUIREMENTS:

1. MATERIALS: CARBON STEEL, COMPOSITIONS 1035 (UNS G10350), 1040 (UNS G10400) AND 1050 (UNS G10500), IN ACCORDANCE WITH QQ-S-635; 1042 (UNS G10420), 1137 (UNS G11370), IN ACCORDANCE WITH ASTM A29, COMPOSITION 11L37 (UNS G11374) IN ACCORDANCE WITH ASM 5020. ALLOY STEEL, GRADES 4130 (UNS G41300), 4340 (UNS G43400), (D) 8740 (UNS G87400), CARBON STEEL, C1110 (UNS G11100) AND C1018 (UNS G10180), IN ACCORDANCE WITH ASTM A29.

   INSERT-NON-METALLIC, NYLON OR EQUIVALENT.

2. FINISH: CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 2.

3. DIMENSIONING AND TOLERANCING: DIMENSIONING AND TOLERANCING SHALL BE IN ACCORDANCE WITH ANSI Y14.5M.

4. HARDNESS: 35 HRC, MAX.

5. THREADS: THREADS IN ACCORDANCE WITH MIL-S-8879.

6. SURFACE TEXTURE: SURFACE TEXTURE, UNLESS OTHERWISE SPECIFIED, SHALL NOT EXCEED 125 MICROINCHES, IN ACCORDANCE WITH ANSI/ASME B46.1.

7. VIBRATION LIFE: MINIMUM VIBRATION LIFE REQUIREMENT SHALL BE THREE TIMES AVERAGE VIBRATION LIFE IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.

8. EDGES: BREAK ALL SHARP EDGES AND REMOVE ALL HANGING BURRS AND SLIVERS.

9. PART NUMBERS: THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I.

   EXAMPLE: MS21080-4
   - 4 — DASH NUMBER
   - MS21080 — BASIC MS NUMBER

   MS21080-4 INDICATES: NUT, SELF-LOCKING, PLATE, ONE LUG, NON-METALLIC INSERT, STEEL: 125 KSI Ftu, 250°; .250-28 UNJF-3B; PLAIN HOLES.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.
4. MS21080 SUPERSEDES NAS 1025 (SEE STATUS NOTE).
5. DESIGN AND USAGE LIMITATIONS: THESE NUTS ARE DESIGNED TO DEVELOP THE TENSILE STRENGTH OF BOLTS AND SCREWS WITH AN ULTIMATE TENSILE STRENGTH OF 125 KSI BASED ON THE CROSS SECTION AREA AT THE BASIC ROOT DIAMETER OF THE THREADS. THESE NUTS ARE DESIGNED TO BE USED ON 3A EXTERNAL THREADS. THESE NUTS SHALL BE USED IN ACCORDANCE WITH THE LIMITATIONS OF MS33588. ONLY NUTS FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED ON QPL 25027 SHALL BE USED.

INTERCHANGEABILITY RELATIONSHIP

MS21080 NUTS CAN UNIVERSALLY REPLACE NAS1025 OF LIKE MATERIAL, THREAD SIZE, PLATING LUBRICANTS (DRY FILM OR NON-DRY FILM), RIVET SPACING AND FASTENING METHOD (PLAIN RIVET HOLES, DIMPLED OR COUNTERSUNK RIVET HOLES OR PROJECTION WELDING), BUT THESE NAS1025 NUTS CANNOT UNIVERSALLY REPLACE MS21080 NUTS.

INTERCHANGEABILITY TABLE

CANCELLED PART NUMBERS	SUBSTITUTIVE PART NUMBERS
NAS1025N06	MS21080-06
NAS1025N06K	MS21080-06K
NAS1025N08	MS21080-08
NAS1025N08K	MS21080-08K
NAS1025N3	MS21080-3
NAS1025N3K	MS21080-3K
NAS1025N4	MS21080-4
NAS1025N4K	MS21080-4K
NAS1025N5	MS21080-5
NAS1025N5K	MS21080-5K
NAS1025N6	MS21080-6
NAS1025N6K	MS21080-6K

PREPARING ACTIVITY: DLA-IS
CUSTODIANS: ARMY- AV NAVY- AS
AIR FORCE- 11 DLA-
REVIEW: IS
USER: CR, MI
PROJECT NUMBER: 5310-1856

MILITARY SPECIFICATION SHEET
TITLE
NUT, SELF-LOCKING, PLATE, ONE LUG,NON-METALLIC INSERT STEEL,125 KSI FTU,250°

SPECIFICATION SHEET NUMBER
MS21080 D 20 APR 93
SUPERSEDING
MS21080C 30 SEPT 87 (SEE NOTE 4)
AMSC- N/A FSC 5310

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